Unit II PHOTOCHEMISTRY

Photochemistry deals with the study of chemical reactions which are caused by the absorption of light(photons). The reactions which take place by absorption of light in the visible and ultraviolet region with the wavelength 800nm and 200nm causes the photochemical reactions.

Laws of photochemistry:

1. **Grothus Draper law or First law:** Only that light which is absorbed by a substance can bring about photochemical reaction.
2. **Stark-Einstein's law of photochemical equivalence or Second law:** One molecule of a substance is activated by the absorption of one quantum of radiation in the primary step of a photochemical reaction.

Quanta(φ) : A body emitting radiation or light of a certain wavelength cannot lose energy in any amount. It must do so in packets of definite size called Quanta. Further light energy will be lost in terms of whole number of quanta's like 1Q, 2Q, 3Q etc and fractional quanta like ½ Q , ¼ Q, $1\frac{1}{2}$ Q etc are not permitted.

One quanta of energy is equal to **hν** i.e **E** = **hν**

Where **h** is Planck's constant and **ν** is frequency of light **ν = c/λ.** Where 'c' is velocity of light and '**λ**' is wavelength.

$\mathbf{E = h\nu = h\frac{c}{\lambda}}$

h = 6.62×10^{-27} erg. Sec or **h= 6.62 ×10^{-34}** Joule.sec. **c = 3×10^{10}** cm/sec = **3×10^{8}** m/sec

1 joule = 10^{7} erg and **1Å = 10^{-8} cm = 10 $^{-10}$m**

Einstein: '*The energy absorbed per mole of the reacting substance is called one Einstein.'*

Einstein = E = Nhν = $\frac{Nh\nu}{\lambda}$ where N is Avogadro number. It is equal to 6.23×10^{23}

Problems: 1. Calculate the energy of one photon of light of wavelength 6000 Å.

2. Calculate the energy of an Einstein of light of wavelength 2537 Å.

Quantum Yield or efficiency (Φ):

Definition: It is the ratio of number of molecules reacted in a given time to the number of quanta f light absorbed in the same time.

Quantum yield or efficiency (φ)

$$\varphi = \frac{number\ of\ molecules\ reacted\ in\ a\ given\ time}{number\ of\ quanta\ of\ light\ absorbed\ in\ the\ same\ time}$$

$$\varphi = \frac{number\ of\ moles\ reacted\ in\ a\ given\ time}{number\ of\ Einstein\ of\ light\ absorbed\ in\ the\ same\ time}$$

If the second law is correct, the quantum yield should be unity. However in most of the photochemical reactions it is not equal to unity. This because the products of the primary steps may undergo a series of secondary processes, which alter the net yield.

Generally the photochemical reactions take place in two stages. They are,

1. **Primary process or Photo process:** In primary process only light is absorbed by the species (atom/molecule/ion) giving rise to the formation of an excited species.
2. **Secondary process:** In the secondary process the excited species produced in the primary process react further without the absorption of additional light (photons). The secondary process is of two types.

a) **Photo physical process:** In this process no chemical change takes place. Eg. Phosphorescence and fluorescence.

b) **Photochemical process :** in this process chemical change takes place. Eg: Decomposition of HI into H_2 & I_2.

<u>Reasons for Low quantum yield (φ<1):</u>

1. Excited species may get deactivated before they give product.
2. There may be a secondary reaction in which the original species which are absorbing light are reproduced.
3. The primary photochemical reaction may be reversed.
4. The collision of the excited species with the non-excited species may cause the former to lose their energy.
5. Some species even after absorption may not active enough to undergo reaction.
   Ex: Photochemical combination of H_2 and Br_2.

<u>Example for low quantum yield reaction:</u>

The photochemical combination of H_2 & Br_2 :

When a mixture of hydrogen gas and bromine vapors is exposed to light, it reacts to give HBr. The reaction follows the following steps:

1. $Br_2 + h\nu \rightarrow 2\,Br^{\cdot}$ Primary process

2. $Br^{\cdot} + H_2 \rightarrow HBr + H^{\cdot}$
3. $H^{\cdot} + Br_2 \rightarrow HBr + Br^{\cdot}$
4. $H^{\cdot} + HBr \rightarrow H_2 + Br^{\cdot}$
5. $Br^{\cdot} + Br^{\cdot} \rightarrow Br_2$

} secondary

The reaction (2) is slow and the reaction (3), (4) & (5) depends on reaction (2). Since the reaction (2) is slow , the reactions (3), (4) & (5) also become slow. Therefore most of Br atoms produced in the primary process recombine to give back Br_2 molecules. Thus HBr molecules formed per quanta of radiation absorbed are less. Hence the quantum yield is less.

Reasons for high quantum yield:

1. The activated species may transfer a part of the absorbed energy to the other species which also become active.
2. There may be a series of chain reactions in which a large number of species react.
3. If the secondary reaction is exothermic , the heat of the reaction may activate other species, which may undergo reaction without absorbing additional photons.
4. Excited molecules may react further by their own without the absorption of additional photons.

Example for high quantum yield reaction- The photochemical combination of H_2 & Cl_2

The quantum yield of the photochemical combination of H_2 & Cl_2 reaction varies between 10^4 to 10^6 in the absence of O_2 . The presence of O_2 slows down the rate of the reaction and thus lowers the quantum yield. The high quantum yield is due to the chain reaction. The reaction may be represented as

1. $Cl_2 + h\nu \rightarrow 2\,Cl^{\cdot}$ Primary process
2. $Cl^{\cdot} + H_2 \rightarrow HCl + H^{\cdot}$
3. $H^{\cdot} + Cl_2 \rightarrow HCl + Cl^{\cdot}$

} secondary Process

The steps 2 & 3 repeat to give the chain reaction. The chain is terminated at walls of the reaction vessel. The termination reqaction is $Cl^{\cdot} + Cl^{\cdot} \rightarrow Cl_2$

If O_2 is present , the chain also gets terminated by the reaction $H^{\cdot} + O_2 \rightarrow H_2O$ (radical)

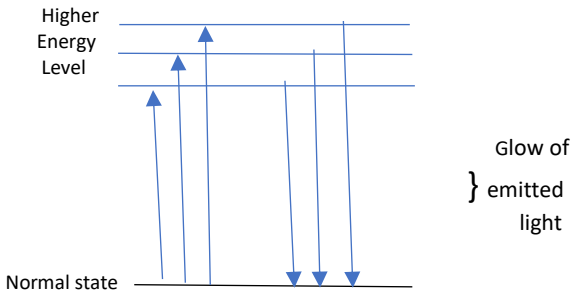
PROBLEMS:

1.A sample of gaseous HI was irradiated by light of wavelength 2537 Å when 307J of energy was found to decompose 1.3×10^{-3} mole of HI. Calculate the quantum yield for the decomposition of HI.

2.In A photochemical reaction of H_2 and Cl_2 using the radiation of wavelength 480 nm, a quantum yield of 10^6 is obtained. What would be the number of moles of HCl produced per mole of radiation energy absorbed?

PHOTOPHYSICAL PROCESS:

Example : Fluorescence: In photo physical process the reacting species absorb light radiation and gets activated. The activated species jump to the higher energy levels. On removal of light source, the activated species return to the ground state instantaneously by emitting their activation energy as a glow of light called fluorescence. Fluorescence is due to the emission of the absorbed light by the reacting species.



Jablonski Diagram of Fluorescence

Definition: The phenomenon of absorption of light radiation at one frequency and re emission of absorbed light in the same or lower frequency is called fluorescence. Fluorescence stops immediately once the source of light is removed . example – substances like fluoresein, CaF_2 ,vapors of I_2,Na, Hg etc.

Fluorescence of two types:

1. **Resonance fluorescence:** if the activated or excited molecules emit radiation of wavelength same as the wavelength of light absorbed, then it is said to exhibit resonance fluoroscence.

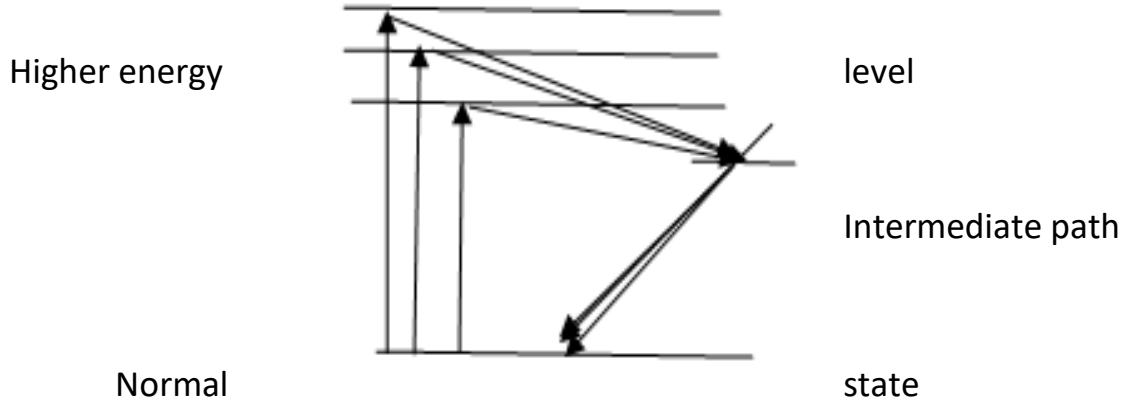
Eg: Hg-vapor wavelength 2337 Å

2. **Sensitized fluorescence:** If the activated or excited molecule emit radiation at shorter wavelength than that of absorbed, then it is said to exhibit sensitized fluorescence.

Fluorescence in liquids depends upon the nature of solvent and the presence of other ions. It increases with its viscosity. Fluorescence in gases increases with decrease of pressure.

Quenching of fluorescence: the fluorescence may be quenched (stopped), when the excited species undergoes collision with a normal molecule before it has the chance to fluoresce. Quenching is due to the transfer of energy from the excited species to the molecules it collides.

Phosphorescence: The phenomenon in which the excited molecule returns to their normal state after a time lag emitting light in lower frequency is called phosphorescence.



In phosphorescence the excited molecules follow a different path for their return to the normal state. Hence there is a time interval between absorption and emission. The substance which show phosphorescence are called phosphorescent substances. Phosphorescence can be seen even after removing the source of light.

Example: Sulphides of alkaline earth metals, ZnS and some dyes when dissolved exhibit phosphorescence.

Photosensitization:

Some chemical reactions do not takes place in the presence of light, but these reactions can be brought in the presence of another substance capable of absorbing light. The substance in the presence of which the photochemical reaction takes place is called photosensitizer and the phenomenon is called photosensitization. Here the photosensitizer molecule absorb light energy, they get activated and these activated molecules exchange energy with the non active molecules t bring up with reaction. The photosensitizer in the process does not undergo any reaction.

Example: Atoms like Cd, Zn and Hg , molecules like SO_2 , benzophenone are used as photosensitizers.

Photosensitization in plants:

In the presence of sun light, CO_2 and H_2O in plants combine together t form carbohydrates. Neither CO_2 nor H_2O absorbs any sun light , but chlorophyll ,present in the green portion of the plant absorbs and acts as a photo sensitizer. Chlorophyll absorbs sun light in the range of 4000 Å -7000 Å and transfers it to CO_2 and H_2O molecules , which in turn react to form a carbohydrate $-(CH_2O)_n$ such as cellulose or sugar. The likely reaction is

$$CO_2 + H_2O + \text{light} \longrightarrow \frac{1}{n}(CH_2O)_n \uparrow + O_2$$

Photo Inhibition:

The substances which can reduce the rate of a photochemical reaction are called photo inhibitors and the phenomenon is called photo inhibition.

Example: In photochemical reaction between hydrogen and chlorine, traces of nitric oxide, sulphur dioxide, oxygen and propylene acts as inhibitors and lowers the quantum yield of HCl.

Chemiluminescence:

In some chemical reactions the energy is liberated in the form of light. This phenomenon of emission of visible light as a result of a chemical reaction is called chemiluminescence. It is reverse of photochemical reactions.

Examples:

1. Oxidation of Phosphorous: phosphorous when exposed to light, gets oxidized and produces visible glow in dark. The reason is that it converts into P_2O_5 in excited state. The reaction is

$4P + 5O_2 \rightarrow P_2O_5^*$

2. Oxidation of 5-aminomophthalic cyclic hydrazide (luminol) by hydrogen peroxide in alkaline solution emits bright green light.

Bioluminescence:

In some biological reactions the energy is liberated in the form of light. The phenomenon of emission of visible light called as cold light as a result of a biological reaction is called bioluminescence.

Example: Cold light emitted by fireflies and jelly fish.

The cold light emitted by fireflies is due to a biological reaction. The fireflies contain a protein luciferin and an enzyme luciferase. This protein and the enzyme in the presence of atmospheric oxygen undergo aerial oxidation to emit cold light. The similar reaction take place in the jelly fish also.

Difference between photochemical and thermal reactions:

Photochemical reaction	Chemical /thermal reaction
1. Reaction takes place by the absorption of light in the visible and ultraviolet region region with the wavelength 800nm-200nm.	1. Reactions takesplace largely in the presence of heat energy
2. The rate of the reaction is very high.	2.The rate of the reaction is low.
3. The activation energy is acquired by the absorption of photons of light associated with a particular wavelength.	3. The activation energy results from the random inter-molecular collisions.
3. Only a single reacting species can be promoted to the excited state.	4.Such selectivity is not permitted.

ACTINOMETRY: An actinometer is an instrument used to measure the intensity of the light absorbed by a chemical substance.

Example: uranyl oxalate actinometer

The uranyl oxalate actinometer consists of a mixture of 0.05M oxalic acid and 0.01 M uranyl sulphate in water. The coloured uranyl ions absorb light (range 2540 Å to 4350 Å) and pass it into oxalic acid molecules. The oxalic acid molecules decompose to give product.

The reaction is, $UO_2^{2+} + h\nu \longrightarrow [UO_2^{2+}]^*$ **(excited)**

$(COOH)_2 + (UO^{2+})^* \longrightarrow UO_2^{2+} \uparrow + CO_2 \uparrow + CO + H_2O$

The amount of oxalic acid reacted is determined by titrating it against standard $KMnO_4$ before and after exposing to light. By knowing the number of quanta of light absorbed at the same duration of time the quantum yield of the reaction can be determined using the equation,

$$\textbf{Quantum yield } (\varphi) = \frac{number\ of\ Oxalic\ acid\ molecules\ reacted\ in\ a\ given\ time}{number\ of\ quanta\ absorbed\ in\ the\ same\ time}$$

Problem: In a uranyl actinometer 5.18×10^{18} molecules under go reaction. During the same time 10.58×10^{18} photons of wavelength 3655 Å were absorbed. Calculate the quantum yield of the reaction.

Ans: Quantum yield (φ) = $\frac{number\ of\ Oxalic\ acid\ molecules\ reacted\ in\ a\ given\ time}{number\ of\ quanta\ absorbed\ in\ the\ same\ time}$

$$= \frac{5.18\times10^{18}}{10.58\times10^{18}} = 0.4896$$

Determination of absorbed intensity:

A photochemical reaction is caused by the absorption of light in the visible / ultra violet region by the reacting species. It is very essential to determine the intensity of the light absorbed to calculate the rate of a photochemical reaction. The schematic diagram of the instrument used for the measurement of light absorbed is shown.

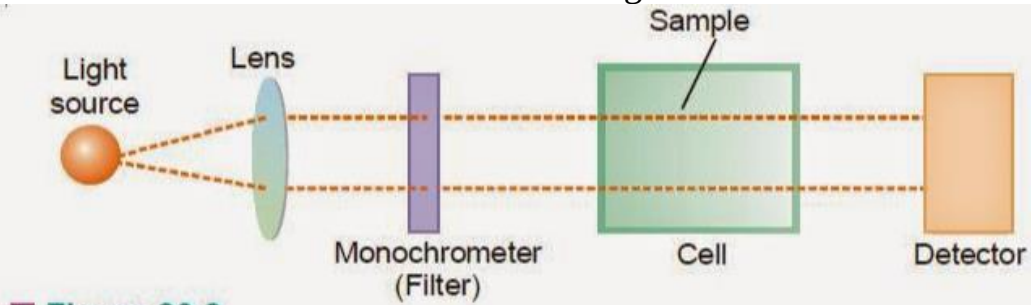


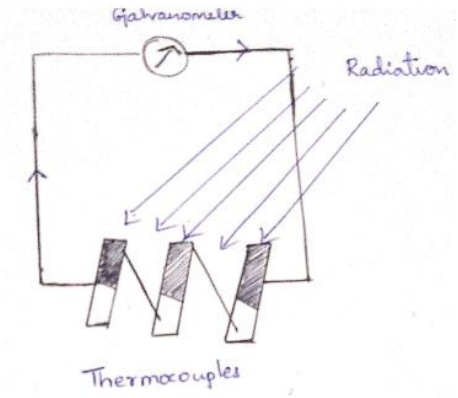
Figure 30.3
Schematic diagram of the apparatus used for measurement of light intensity.

Light beam from a light source like tungsten filament / mercury vapor lamp is first passed through a converging lens and then through a monochromator to produce a monochromatic beam. This monochromatic beam is then made to pass through a quartz cell in which the reaction sample is taken. The part of the light that is transmitted strikes the detector which can be measured. The detectors generally used are thermopile, photoelectric cell or chemical actinometers.

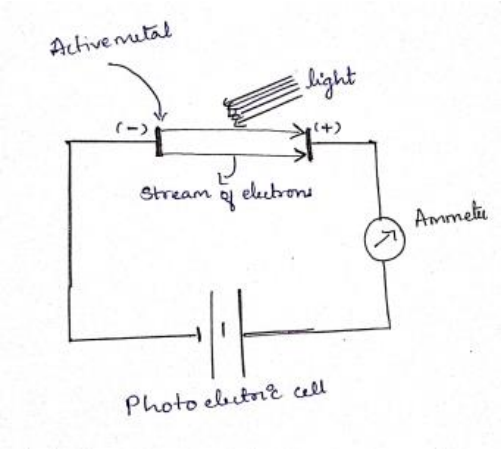
Experimental Part: The intensity of the incident light I_0 is first measured by passing the light through empty cell. The cell is filled with reaction sample and the same beam of light is passed through it. The light striking the detector I_t gives the intensity of the transmitted beam. The absorbed intensity, $I_a = I_0 - I_t$

Detectors: These are the devices used to measure the intensity of transmitted beam. Different types of detectors are,**1) Thermopile 2) Photoelectric cell 3) Chemical actinometer.**

1.Thermopile: The thermopile is made of a series of thermocouples in which metals like bismuth and silver are joined together . the one end of the couple is blackened with lamp black. When the radiation falls on the blackened end, it absorbs the energy and gets heated up. The temperature difference between the blackened end and the other end causes a current to flow in the circuit which is recorded in a galvanometer. The current generated is proportional to the intensity of the transmitted beam. The thermopile is previously calibrated before use. The circuit diagram of a thermopile is shown in the figure.



2.Photoelectric cell: The photoelectric cell used to measure the intensity of light is shown in the diagram. It consists of metal electrodes made of active metals like cesium,sodium or potassium. They are connected to an ammeter to measure the current.The transmitted beam of radiation is made to fall on the active metal electrodes, which intern emits electrons. A current flows through the circuit which can be measured with an ammeter. The intensity of the light falling on the electrodes is proportional to the current generated.



3.Chemical actinometer :- Ex: Uranyl actinometer

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